Number 5 — Calculators · Estimating · Upper and lower bounds

Calculator allowed except in question 4. Marks in brackets. ★ = harder question.

1 Calculate, giving each answer to 3 s.f.: [4]

a $2.4 + 2.4^2 + 2.4^3$ **b** $\dfrac{2.6 \times 10^3}{2.6^3}$ **c** $\dfrac{\sqrt{230}}{5.1 \times 10^3 - 3.8 \times 10^2}$

d $\dfrac{\sqrt{8.1 \times 10^3}}{\sqrt{2.5 \times 10^{-5}}}$

2 Write the lower and upper bounds of each measurement. [8]

a 34 m (nearest metre) **b** 62 kg (nearest kg) **c** 480 s (nearest 10 s)

d 2300 km (nearest 100 km)

3 $w = \dfrac{x + y}{z}$ where $x = 3.5 \pm 0.5$, $y = 2 \pm 0.5$ and $z = 0.8 \pm 0.1$. Find the lower and upper bounds of w. [4]

4 Without a calculator, estimate each value in standard form to 1 s.f. Show your rounding. [4]

a $2.1 \times 10^3 \times 3.9 \times 10^2$ **b** $\dfrac{6.1 \times 10^4}{2.9 \times 10^2}$ **c** $\sqrt{8.9 \times 10^4}$ **d** $\dfrac{5.1\pi \times 10^3}{\sqrt{9.3}}$

5 A runner covers 400 m, measured to the nearest 10 m, in 52 s, measured to the nearest second. Find the lower and upper bounds of her average speed in m/s. [3]

6 The area of a semicircle is measured as 60 m² to the nearest 5 m². Find the lower and upper bounds of its perimeter. (Area of a circle πr^2, circumference $2\pi r$.) [2]

Total: 25 marks

Check: bounds are ± half the unit of rounding, lower bound included, upper not · for a quotient, max = max ÷ min and min = min ÷ max · estimate by rounding to 1 s.f. before calculating · brackets round numerators and denominators on the calculator.